

Dez Calimese — Blockchain Security Engineer

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Summary

With 5 years in blockchain engineering and founding roles at early-stage companies, I build and audit production-grade DeFi infrastructure across EVM and Solana. My work spans ERC4626 vault architectures, multi-chain AI agent systems, perpetual futures trading platforms on Hyperliquid, and multi-chain transaction infrastructure using threshold cryptography. I recently completed the Rektoff x Solana Foundation Rust Security Bootcamp, adding formal Solana/Anchor audit methodology to a background that already includes smart contract security, full-stack development, and taking products from zero to one under ambiguity.

Key Achievements

- Founding Engineer at Omo Protocol, leading smart contract development for standardized ERC4626 vaults, frontend engineering, and advanced AI agent integrations—contributions that helped secure over \$1M in pre-seed funding.
- Principal UI/UX Engineer at Bricks Exchange, managing a 4-person dev team through reviews, mentorship, and deployment oversight.
- CTO at Minting Melodies, driving end-to-end blockchain application development, test automation, and infrastructure reliability.

Technical Skills

- **Languages:** Solidity, TypeScript, Python, Rust, Move, Cairo, SQL
- **Smart Contracts & Security:** Foundry (fuzzing/invariant testing), Hardhat, EVM internals, gas optimization, MEV protection, threat modeling, security audits, formal verification concepts
- **Multi-Chain & Infrastructure:** Threshold cryptography (Lit Protocol), cross-chain messaging, bridging protocols, off-chain relayers, RPC optimization, transaction simulation
- **Protocols & Tools:** ERC4626 vaults, Uniswap, Morpho, Hyperliquid, Tenderly, Subgraphs, Docker, AWS, Linux
- **Frontend & Development:** Next.js, React, React Native, Wagmi, Viem, web3.js, TailwindCSS, Framer, Figma
- **Security Analysis:** Slither, Echidna, Foundry fuzz testing, Mythril, Tenderly testnets, CI/CD with GitHub Actions

Experience

Omo Protocol • Remote
Founding Engineer

11/2023 — 11/2025

- Architected and implemented ERC4626 standardized vaults using Solidity and Foundry, and contributed to a trading platform for spot and perpetual futures on Hyperliquid with integrated DeFi vaults built with Next.js, Tailwind CSS, and TypeScript
- Developed a multi-chain transactions plugin for ElizaOS leveraging Lit Protocol's Programmable Key Pairs and threshold cryptography, enabling secure decentralized key management for AI agents

- Developed Python scripts leveraging Coinbase MPC Wallets to enable AI agents to execute DeFi strategies across Morpho, Uniswap, and Omo's native contracts
- Authored Rust integrations using the Nucleus SDK to enable AI agents on Hyperliquid to interact with on-chain primitives including AMMs, liquid staking, and lending protocols
- Implemented Tenderly's Virtual Testnet framework and testing methodologies, resulting in a 50% increase in testing efficiency and improved contract reliability

Bricks Exchange • Remote
Lead Blockchain Developer

11/2022 — 11/2024

- Lead a team of four frontend developers as the principal engineer responsible for UI/UX development
- Onboarded and trained three junior frontend developers, providing code reviews and mentoring to improve team skills, resulting in 25% faster delivery timelines
- Collaborated cross-functionally with backend engineers, smart contract developers, and security analysts to design, code, and thoroughly test numerous blockchain applications, emphasizing clear communication and relationship building
- Developed project plans and timelines for frontend development initiatives using Next.js, React Native for Web, TypeScript, JavaScript, Wagmi, Viem, and TailwindCSS, focusing on delivering intuitive and appealing UIs within established deadlines
- Leveraged agile methodologies like Scrum and Kanban using project management tools like ClickUp to plan, track progress, and collaborate through all phases of the software development life cycle from conception to deployment

Minting Melodies • Remote
Chief Technology Officer

08/2022 — 02/2023

- Provided technology leadership and support through the development of smart contracts, landing pages, and internal infrastructure to empower artists in creating NFT collections and streamlining royalty distribution
- Optimized royalty distribution through smart contracts, automating 80% of the process
- Established test automation pipelines to support debugging and issue resolution, emphasizing thorough testing and review
- Contributed high-quality code through rigorous processes, remaining focused on meeting project timelines and objectives
- Stayed current with emerging blockchain protocols and assets to inform effective smart contract deployment, development, and testing

Open-Source Contributions

ElizaOS • Lit Plugin • GitHub Link

01/2025

- Developed and contributed a multi-chain transactions plugin leveraging Lit Protocol's Programmable Key Pairs and threshold cryptography, enabling secure decentralized key management for AI agents across networks.
- Built as part of Omo Protocol's AI agent infrastructure; merged into the open-source ElizaOS framework.

Security Research

MetaLend Security Audit • *Rektoff x Solana Foundation Bootcamp Capstone Project* 2026
Full report available upon request.

- Audited a Solana lending protocol, identifying high-severity vulnerabilities across oracle integrity, borrow, withdraw, and liquidation instructions.
- Delivered findings with working Anchor PoCs confirming exploitability.

Selected Projects

Tessera • *Next.js, Gemini 3.1 Pro, Sigma.js, Supabase* • Cerebral Valley x Vercel x DeepMind Hackathon, NYC • [GitHub Link](#) 04/2026

- Built a solo multimodal AI agent that ingests content from URLs, screenshots, YouTube videos, and live market feeds, extracts entities and relationships via Gemini 3.1 Pro, and renders them as an interactive force-directed knowledge graph with AI-synthesized summaries per node
- Migrated from react-force-graph-2d to Sigma.js with WebGL rendering mid-build to support graph scalability beyond 2,000 nodes, and implemented a local-first architecture using IndexedDB as the primary data store with Supabase as cloud backup for offline functionality and instant load times
- Resolved integration issues with Google Search grounding by swapping from the @ai-sdk/google wrapper to the @google/genai SDK directly, and adapted to breaking changes in AI SDK v6's streaming response and chat submission APIs during the event

Staking Application • *Solidity, Hardhat, Scaffold-Eth* • [GitHub Link](#) 07/2022 — 08/2022

- Full-stack decentralized staking app with deposit, withdrawal, and interest accrual.

RSVP Dapp • *Solidity, Next.js, GraphQL, Web3Storage* • [GitHub Link](#) 07/2022 — 08/2022

- Decentralized event creation and RSVP platform with on-chain data storage.

Multi-Signature Wallet • *Solidity, Truffle, React* • [GitHub Link](#) 09/2021 — 01/2022

- Secure multi-sig transfer system with approval workflows from listed accounts. Built during ConsenSys Bootcamp.

Decentralized Exchange • *Solidity, Truffle, React, Bootstrap* • [GitHub Link](#) 05/2021 — 06/2021

- Order-book DEX enabling custom ERC-20 token trading with on-chain settlement.

Security Training & Certifications

- **Rektoff x Solana Foundation Rust Security Bootcamp** — Rust internals, Solana/Anchor security, capstone lending protocol audit with working PoCs
- **RareSkills Advanced Solidity Bootcamp** — Gas optimization, EVM internals, security patterns
- **Secureum Smart Contract Security Bootcamp** — Advanced vulnerability patterns, formal verification concepts, audit methodologies
- **Encode Club Bootcamps** — Expert Solidity, Move (Aptos/Sui), Cairo/StarkNet
- **ConsenSys Academy Ethereum Smart Contract Developer Bootcamp**
- **Ackee Blockchain Solana Auditors Bootcamp**

Education

Cardinal McCarrick High School • South Amboy, NJ
HS Diploma